Organic acids and details

Acid: $\underbrace{HA}_{\text{acid}} + H_2O \underset{}{\overset{Ka_1}{\rightleftharpoons}} H_3\overset{+}{O} + \underbrace{A^-}_{\text{conjugate base.}}$

$$Ka = \frac{[H_3^+O][A^-]}{[HA]} \quad ; \quad \underbrace{pKa}_{} = -\log Ka$$

pka

lower pka — acidic

higher pka — basic

Factors on which acid strength depends:

1. Strength of H-A bond
2. Stability of conjugate base $[A^-]$
3. Electronegativity of A
4. Solvation of A^- in presence of solvent.

strong acid	moderate	weak	extremely weak.
pKa < 1	1 – 5	5 – 15	> 15
trifluoroacetic acid	carboxylic acid.	alcohols phenols.	

0.23

pKa = −7

HCl mineral acid

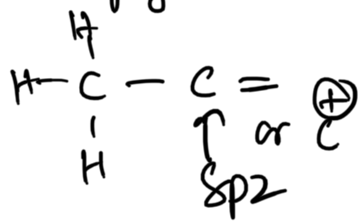
benzoic acid 

pKa = 4.19

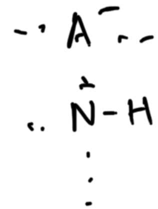
CH_3COOH, pka = 4.76

Structure of acid

- Inductive effect
  - $-I$
  - $+I$
- Mesomeric effect
  - $-M$ | $-R$
  - $+M$ | $+R$
  - $+M$ lp of e^-s
  - $\rightarrow \overset{+}{N}=O$ $-M$
  - $-O-$
- Hyper conjugation



- H-bonding



- Steric effects.

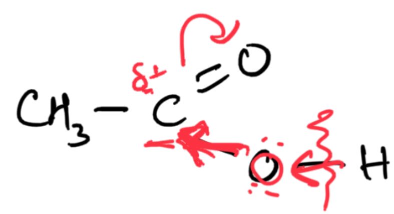
ALIPHATIC ACIDS

$CH_3 \rightarrow COOH$ < $\underbrace{CH_3\,CH_2}_{+I} \rightarrow COOH$ < $(CH_3)_3C \rightarrow COOH$

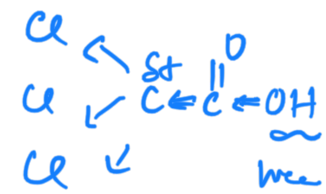
$+I$

pKa: 4.76 | 4.88 | 5.05

stronger | | weak acid



	$F \leftarrow CH_2COOH$	$Cl-CH_2COOH$	$Cl_3-C-\overset{O}{\overset{\|}{C}}-OH$
pKa	2.57	2.86	0.65
	II	III	I
		weakest	strongest

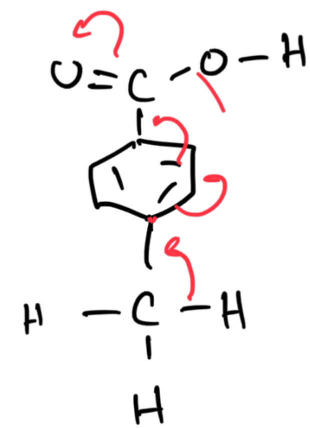


Aromatic carboxylic acids.

	pKa
C_6H_5COOH	4.20
$m-CH_3C_6H_4COOH$	4.24
$p-CH_3-C_6H_4COOH$	4.34

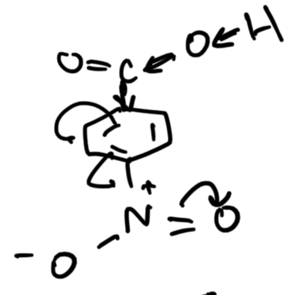
most acidic

least acidic



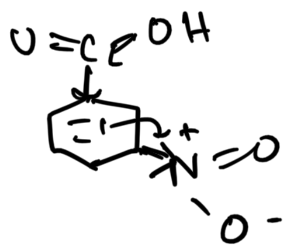
	pKa	
o-NO_2 C_6H_4COOH	2.17	— strongest
m-NO_2 C_6H_4COOH	3.45	— weakest
p-NO_2 C_6H_4COOH	3.43	

NO_2 : $-I$, $-M$ / o, p-

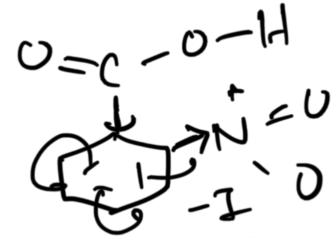


para $-M$

weak $-I$



$-I$ effect



$-I$

$-M$

$-I$

	pKa		
o-Cl: C_6H_4COOH	2.94	most acidic	−I > +M ortho
m-Cl: C_6H_4COOH	3.83		−I, no +M
p-Cl: C_6H_4COOH	3.99.	least acidic	−I < +M

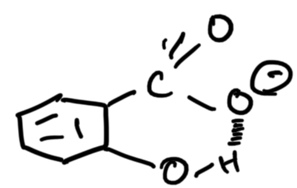
−I +M

EN lps

o-OH C_6H_4COOH	2.98	most acidic,	−I > +M
m-OH C_6H_4COOH	4.08		−I, no +M
p-OH C_6H_4COOH	4.58	weaker acid.	+M > −I

−:OH

−I, +M



CB stabilized

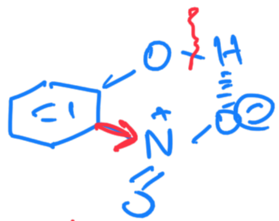
Phenols

	pKa
C_6H_5OH	9.95
o - NO_2 C_6H_4OH	7.23
m - NO_2 C_6H_4OH	8.35
p - NO_2 C_6H_4OH	7.14 — most acidic

NO_2

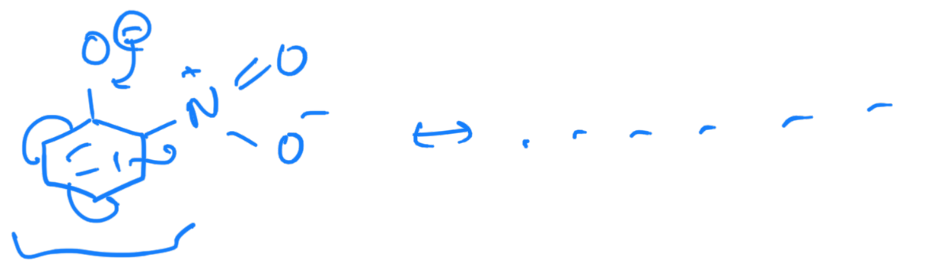
-I

-M



-M

-I



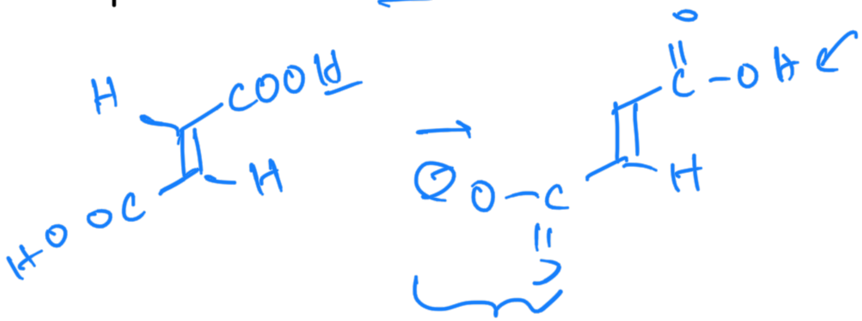
Stability of CB.

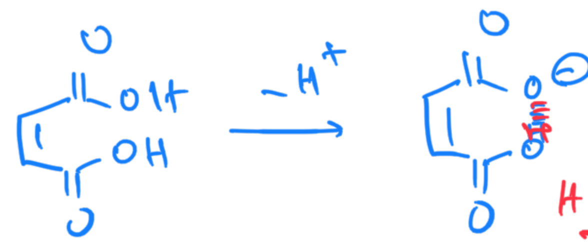
Oxalic acid > malic acid > succinic acid

$HOOC-COOH$ $\quad$ $HOOC \leftarrow CH_2 \rightarrow COOH$ $\quad$ $HOOC \leftarrow \underbrace{CH_2-CH_2}_{4.19} \rightarrow COOH$

1.23 $\quad$ 2.83

	fumaric (trans-)	maleic acid (cis-)
pKa_1	3.02	1.92
pKa_2	4.38	6.23





H-bonding

CB stable

Organic bases

$$\underbrace{:B}_{\text{Base}} + H_2O \underset{}{\overset{K_b}{\rightleftharpoons}} \underbrace{BH^+}_{\text{conjugate acid}} + OH^-$$

$$K_b = \frac{[BH^+][OH^-]}{[B:]}$$

$$BH^+ + H_2O \overset{K_a}{\rightleftharpoons} :B + H_3O^+$$

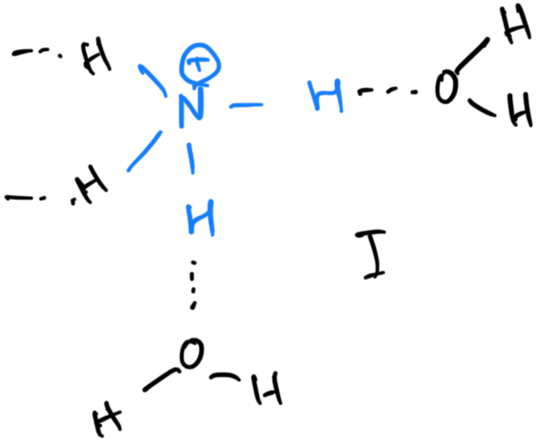
$$K_a = \frac{[B][H_3O^+]}{[BH^+]}$$

strong base : weak CA
strong acid : weak CB

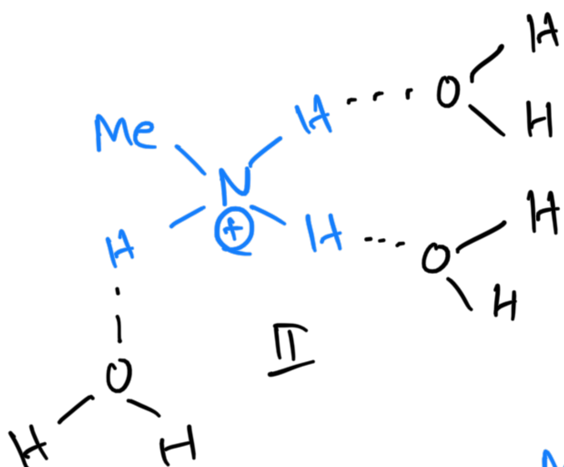
factors on which basicity depends

1. Stability of CA
2. lp of electrons
3. solvation of CA by polar solvent.

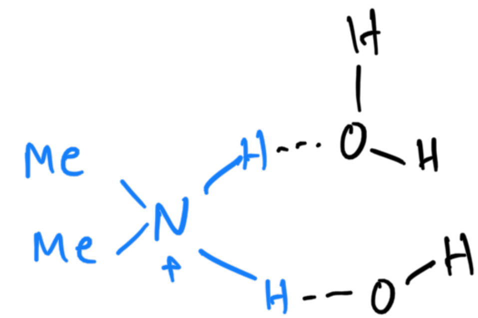
	$:NH_3$	$MeNH_2$	Me_2NH	Me_3N
	I	II	III	IV
pKa (H_2O)	9.25	10.64	10.77	9.80
	(least basic)		(most basic)	



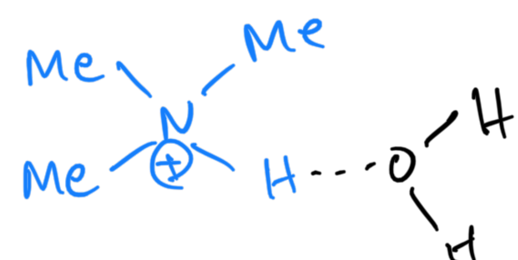
I



II

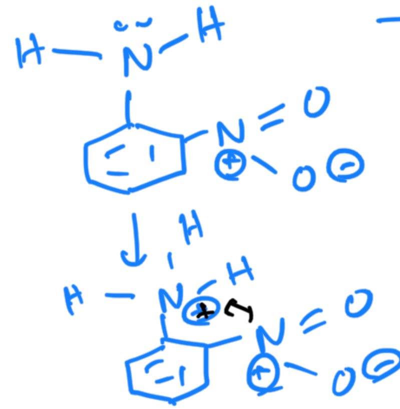


III



IV

	pKa	
o - NO_2 - $C_6H_4NH_2$	$\underline{-0.28}$	I
m - NO_2 - $C_6H_4NH_2$	2.45	II
p - NO_2 - $C_6H_4NH_2$	0.98	III



→ least basic

→ most basic

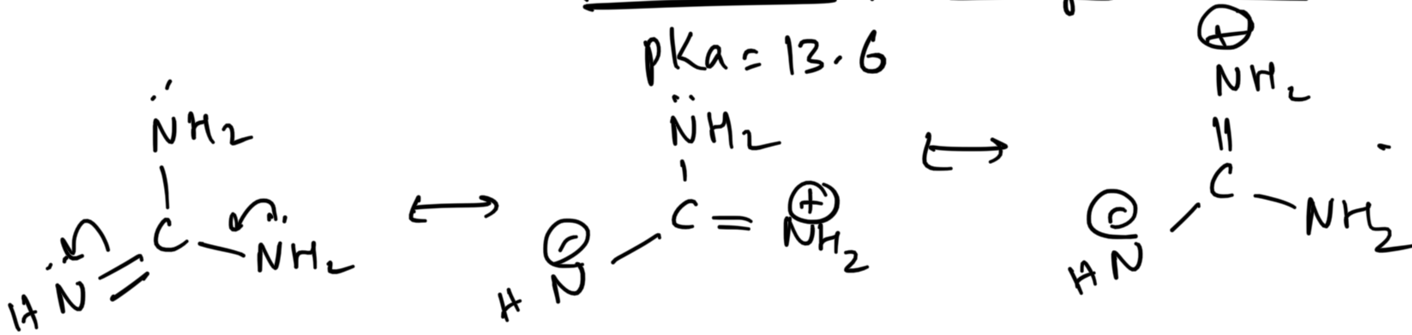
CA

	pKa	
o - MeO - $C_6H_4NH_2$	$\underline{4.49}$	-I, +M
m - MeO - $C_6H_4NH_2$	4.20	
p - $\underbrace{\text{MeO}}_{+M}$ - $C_6H_4NH_2$	$\underbrace{5.29}_{\text{Most basic}}$	+M, no -I

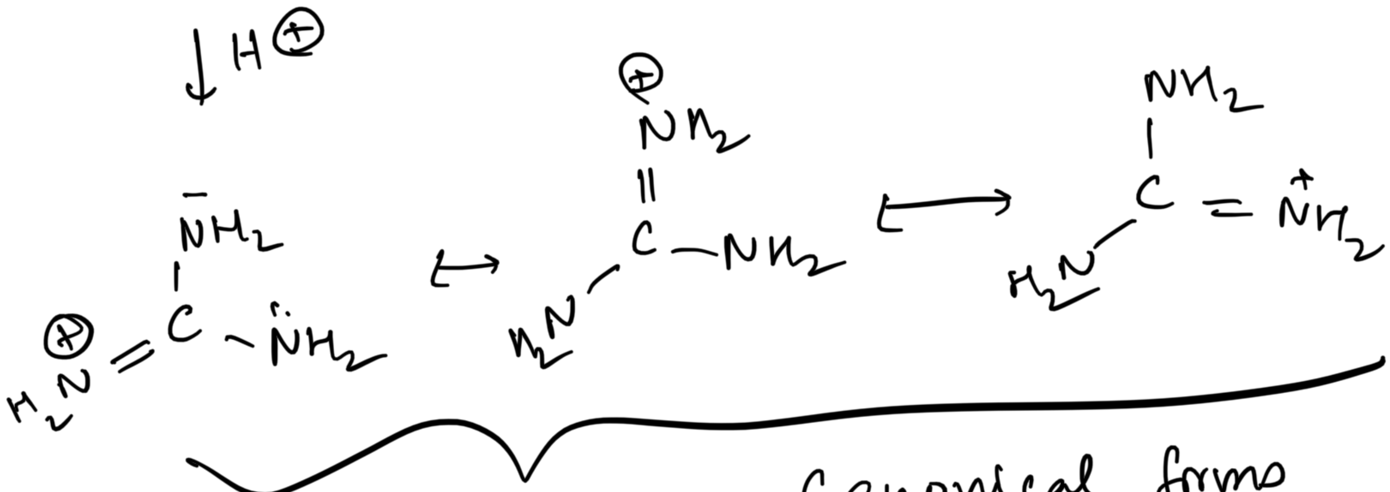
	pKa
o - OH - $C_6H_4NH_2$	$\underline{4.72}$
m - OH - $C_6H_4NH_2$	4.17
p - $\underbrace{\text{OH}}_{+M}$ - $C_6H_4NH_2$	$\underbrace{5.30}_{\text{Most basic}}$

Guanidine : Strongest base

pKa = 13.6



Neutral molecule



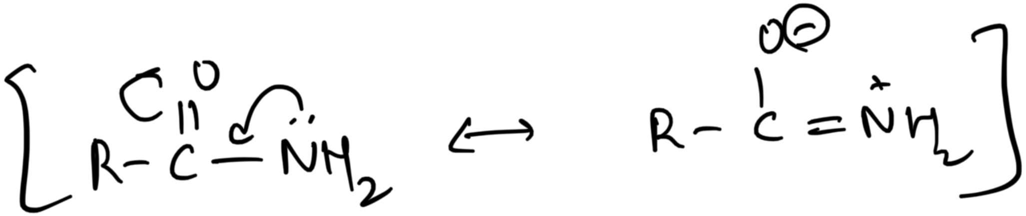
CA

Canonical forms

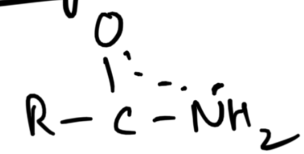
$H_3\ddot{N}$ > $R-CH=\ddot{N}H$ > $R-C\equiv\ddot{N}$

sp^3 sp^2 sp

Order of basicity



weakly basic



pyrrole

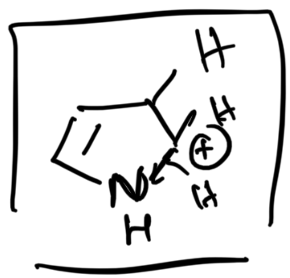
pka -0.27

acidic

$6\pi e^-$

pyridine

5.21



not stable

CA